A simulation exercise

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Overview

We set lambda 0.2, We drew 40 random exponential values and calculated their mean and repeated the process 1000 times.

- For 1, Sample Mean Vs Theoretical Mean - we compared the mean of 1000 simulated means against the theoretical mean.
- For 2, Sample Variance vs Theoretical Variance - we calculated variance and standard deviation of our sample means and compared standard deviation against theoretical standard deviation on plot.
- For 3, We evaluated if the distribution of sample means forms a bell curve.

1.Show the sample mean and compare it to the theoretical mean of the distribution.

```
library(ggplot2)
lambda <- 0.2
n <- 40
sims <- 1000
rexp(40,0.2)
```

```
##  [1]  0.9924714 16.1605272  1.8931400 15.0195525  5.2591420 10.8873033
##  [7]  2.5467360  6.5136638  4.8303034  2.0201288  1.4382931  7.9176452
## [13]  9.9595827 19.3127473  0.1484801 16.5768276  5.2121951 14.5512744
## [19]  1.2743439 27.3601800  5.6563361  1.0474287  1.7213048  1.7045896
## [25]  0.5259742  6.8578999  2.7787202  1.3955093 14.5245259 18.9140363
## [31]  2.2236483  7.7988064  5.6182267  2.5370571 14.4012520  1.0734679
## [37] 17.5319120  1.2947429  2.2694700  6.0950344
```

```
mean(rexp(40,0.2))
```

```
## [1] 4.30968
```

```
set.seed(44)
sim_means <- replicate(1000, mean(rexp(40, 0.2)))
```

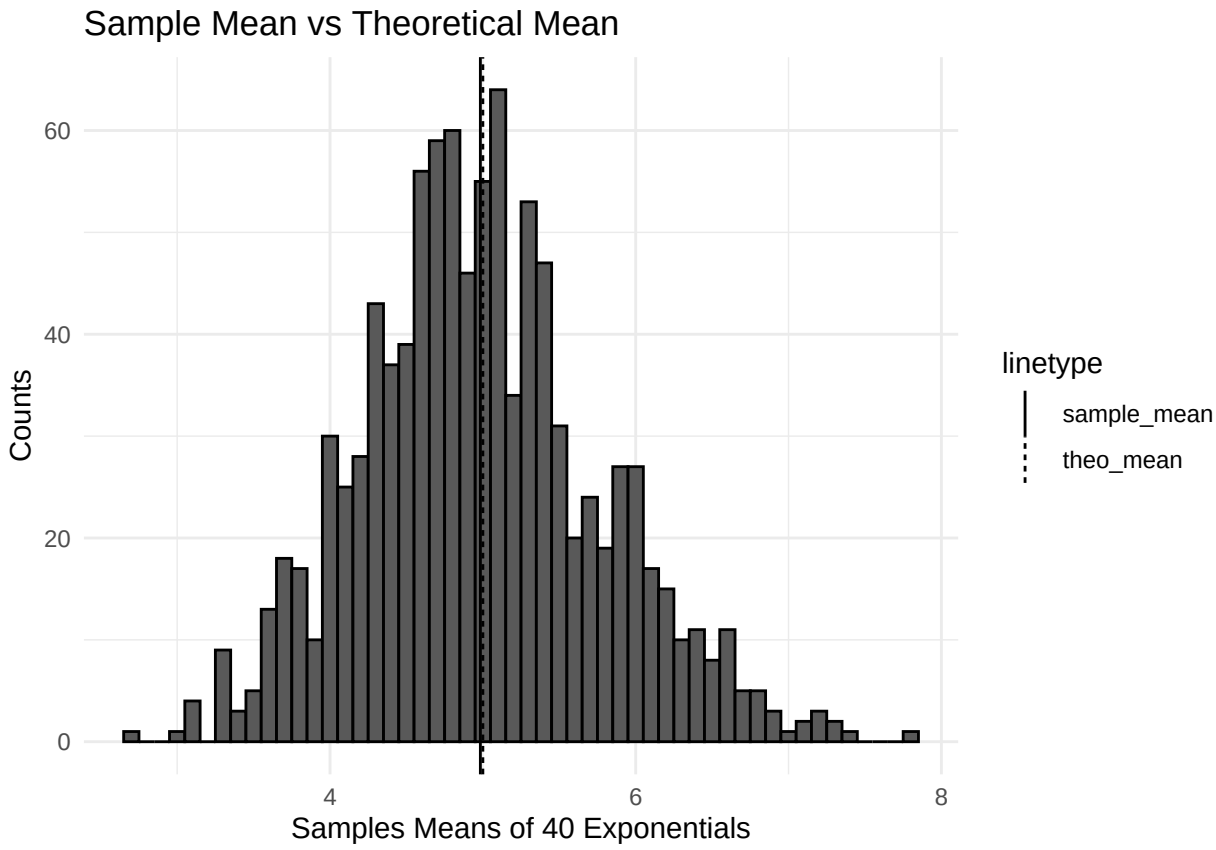
```
sample_mean <- mean(sim_means)
theo_mean <- 1 / 0.2
sample_mean; theo_mean
```

```
## [1] 4.983733
```

```
## [1] 5
```

The sample mean is 4.98, - very similar to theoretical mean - 5.

```
mydata <- data.frame(sim_means)
ggplot(mydata,
       aes( x = sim_means)) +
        geom_histogram(binwidth = 0.1, color = "black") +
        geom_vline(aes(xintercept = sample_mean, linetype = "sample_mean")) +
        geom_vline(aes(xintercept = theo_mean, linetype = "theo_mean")) +
        labs(title = "Sample Mean vs Theoretical Mean",
             x = "Samples Means of 40 Exponentials",
             y = "Counts") +
        theme_minimal()
```



We can see the solid line - sample mean and the dash line - theoretical mean are almost on the same line. So, we can see they are almost the same.

2.Show how variable the sample is (via variance) and compare it to the theoretical variance of the distribution.

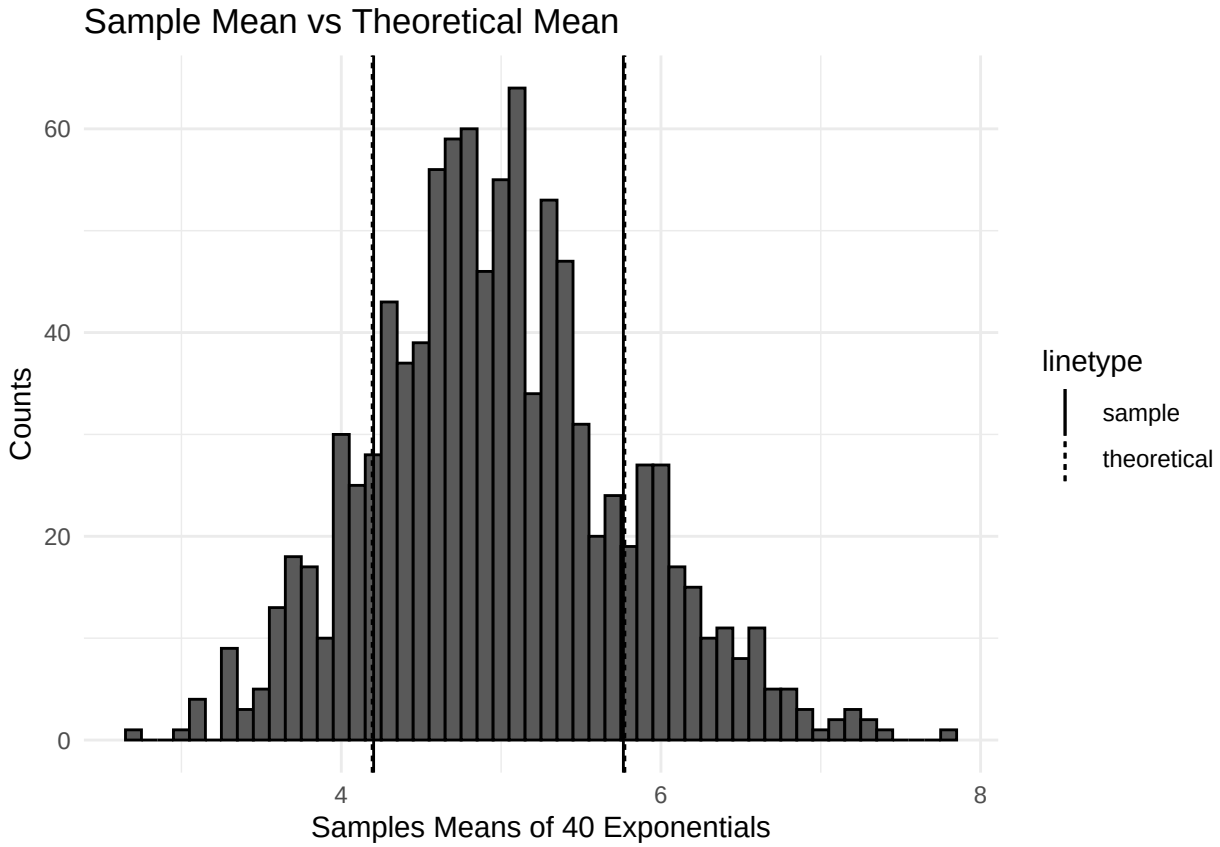
```
sample_var <- var(sim_means)
theo_var <- (1/lambda)^2/n
sample_var;theo_var
```

```
## [1] 0.6104049
```

```
## [1] 0.625
```

```
sample_sd <- sd(sim_means)
theo_sd <- sqrt(theo_var)
```

```r
ggplot(mydata,
       aes( x = sim_means)) +
        geom_histogram(binwidth = 0.1, color = "black") +
        geom_vline(aes(xintercept = sample_mean + sample_sd, linetype = "sample")) +
        geom_vline(aes(xintercept = sample_mean - sample_sd, linetype = "sample")) +
        geom_vline(aes(xintercept = sample_mean + theo_sd, linetype = "theoretical")) +
        geom_vline(aes(xintercept = sample_mean - theo_sd, linetype = "theoretical")) +
        labs(title = "Sample Mean vs Theoretical Mean",
             x = "Samples Means of 40 Exponentials",
             y = "Counts") +
        theme_minimal()
```

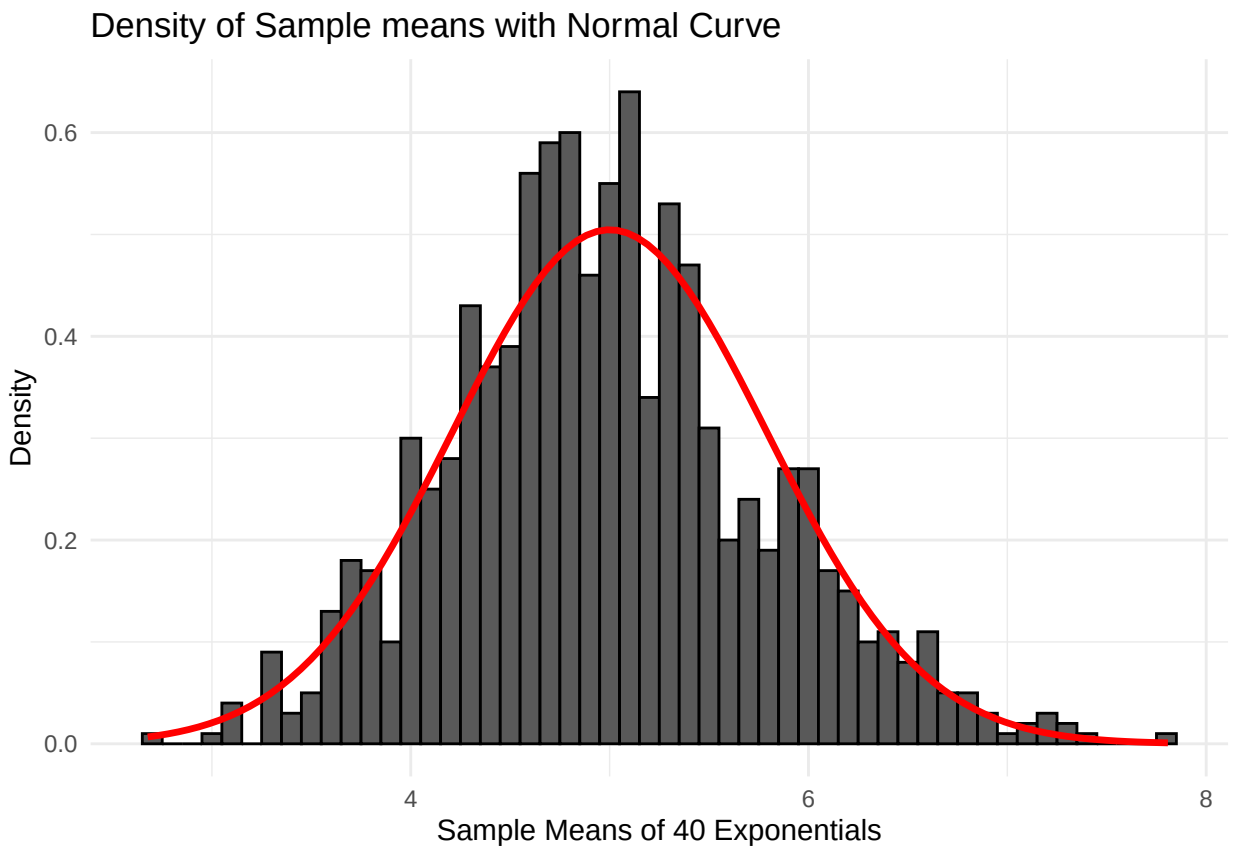


The sample variance is 0.6104049, - very similar to theoretical variance - 0.625. On the graph, it shows + or - 1 sd from sample mean. Solid lines are for + or - sample standard deviation from sample mean while the dash lines for + or - theoretical standard deviation from sample mean. We can see these line overlap as they are the same.

3.Show that the distribution is approximately normal.

```r
ggplot(mydata,
       aes( x = sim_means)) +
        geom_histogram(binwidth = 0.1, color = "black", aes(y = after_stat(density))) +
        stat_function(fun = dnorm, args = list(mean = theo_mean, sd = theo_sd), color = "red", linewidth
        labs(title = "Density of Sample means with Normal Curve",
             x = "Sample Means of 40 Exponentials",
             y = "Density") +
```

```
  theme_minimal()
```



In the graph, the red line - theoretical line show how the sample means form a bell curve.